
Face Generation Using Deep Generative Models

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Abstract

This project investigates face generation using a range of deep generative models, including Autoencoders (AE), Variational Autoencoders (VAE), Generative Adversarial Networks (GAN), Wasserstein GAN (WGAN), and Denoising Diffusion Probabilistic Models (DDPM). These models are trained on the CelebA dataset to learn meaningful latent representations and synthesize realistic human faces. The study focuses on comparing these models in terms of reconstruction quality, diversity of generated samples, and training stability. While Autoencoders provide a strong baseline for reconstruction, Variational Autoencoders introduce probabilistic modeling of the latent space, enabling controlled sampling. GAN-based models significantly improve perceptual realism through adversarial training, whereas diffusion models achieve state-of-the-art image quality with highly stable optimization. The report presents architectural details, experimental observations, generated outputs, and training dynamics to highlight the strengths and limitations of each approach.

1. Introduction

Face generation is a fundamental problem in computer vision that involves modeling the distribution of high-dimensional image data. The objective is to generate realistic facial images that closely resemble samples from a given dataset while maintaining diversity across outputs. This task is inherently challenging due to the complexity of human facial structures, variations in pose, lighting conditions, and identity-specific features.

Recent advancements in deep learning have led to the development of powerful generative models. This project systematically explores the evolution of generative models, beginning with deterministic approaches such as Autoencoders and extending to probabilistic models like VAEs, adversarial frameworks such as GANs and WGANs, and finally diffusion-based generative processes. The goal is to understand how each model improves upon its predecessor and to analyze their trade-offs in terms of visual quality, diversity, and training behavior.

2. Literature Review

2.1 Autoencoders

Autoencoders are neural networks designed to learn compact representations of input data through an encoder-decoder architecture. The encoder maps an input image to a lower-dimensional latent representation, while the decoder reconstructs the image from this representation. Early work by Hinton and Salakhutdinov^[1] demonstrated the effectiveness of deep autoencoders for dimensionality reduction. While Autoencoders are capable of capturing the global structure of images, they are inherently limited in generative capability because they learn deterministic mappings. As a result, they tend to reconstruct training samples rather than generate diverse new outputs.

2.2 Variational Autoencoders

Variational Autoencoders (VAEs), introduced by Kingma and Welling^[2], extend traditional Autoencoders by incorporating a probabilistic latent space. Instead of mapping inputs to fixed latent vectors, the encoder predicts parameters of a distribution, typically a Gaussian characterized by mean and variance. Sampling from this latent distribution enables the generation of new data points. However, VAEs often produce blurry images due to the combination of reconstruction loss and Kullback–Leibler (KL) divergence, which enforces smoothness in the latent space and leads to averaging over multiple plausible outputs.

2.3 Generative Adversarial Networks

Generative Adversarial Networks (GANs), proposed by Goodfellow et al.^[3], introduced a novel framework for generative modeling using adversarial training. A generator network learns to produce realistic images from random noise, while a discriminator network attempts to distinguish between real and generated samples. This minimax game encourages the generator to produce highly realistic outputs. Subsequent works, such as DCGAN (Radford et al.^[4]), demonstrated the effectiveness of convolutional architectures in stabilizing GAN training for image generation tasks. Despite their success, GANs are prone to issues such as mode collapse, non-convergence, and sensitivity to hyperparameters.

2.4 Wasserstein GAN

Wasserstein GAN (WGAN), proposed by Arjovsky et al.^[5], addresses the instability of traditional GANs by replacing the Jensen-Shannon divergence with the Wasserstein distance (Earth Mover's Distance). This formulation provides smoother gradients and a more meaningful loss metric, improving training stability. The introduction of weight clipping, and later gradient penalty (Gulrajani et al.^[6]), further enhanced convergence behavior. WGAN significantly

reduces issues such as mode collapse and provides more stable learning dynamics compared to standard GANs.

2.5 Diffusion Models

Diffusion models represent a recent advancement in generative modeling. Ho et al. [\[7\]](#) introduced Denoising Diffusion Probabilistic Models (DDPM), which generate images by reversing a gradual noise addition process. In the forward process, noise is incrementally added to the data until it becomes pure noise, while in the reverse process, a neural network learns to denoise the data step-by-step. This iterative approach enables diffusion models to generate highly realistic and detailed images. Subsequent improvements, such as those by Nichol and Dhariwal [\[8\]](#), have further enhanced efficiency and quality, making diffusion models state-of-the-art for image synthesis tasks.

3. Methodology

3.1 Dataset

The CelebA dataset is used for training all models in this project. It contains over 200,000 images of celebrity faces with significant diversity in terms of facial attributes, lighting conditions, and backgrounds. The dataset provides a rich distribution for learning realistic face representations.

All images are preprocessed before training. This includes resizing them to a fixed resolution, typically 64×64 pixels, to ensure uniform input dimensions across models. The pixel values are normalized to the range $[-1, 1]$ to facilitate stable training. In some cases, face alignment techniques may be applied to improve consistency across samples.

3.2 Model Architectures

The Autoencoder architecture consists of an encoder network that compresses the input image into a latent representation and a decoder network that reconstructs the image from this representation. The model is trained using a reconstruction loss, typically mean squared error, which measures the difference between the input and reconstructed images.

The Variational Autoencoder modifies this architecture by introducing a probabilistic latent space. The encoder predicts the mean and variance of a Gaussian distribution, and latent vectors are sampled using the reparameterization trick. The loss function combines reconstruction loss with a KL divergence term, which regularizes the latent space to follow a standard normal distribution.

In GAN and WGAN architectures, the generator learns to map random noise vectors to realistic images, while the discriminator or critic evaluates their authenticity. The adversarial training setup pushes the generator to produce increasingly realistic outputs. WGAN improves this process by using a Wasserstein loss, which provides more stable gradients and better convergence.

Diffusion models follow a two-stage process. In the forward process, noise is gradually added to the input images over multiple steps until the data becomes pure noise. In the reverse process, a neural network is trained to predict and remove this noise step-by-step, eventually reconstructing a realistic image. This formulation leads to highly stable training and superior image quality.

4. Experiments

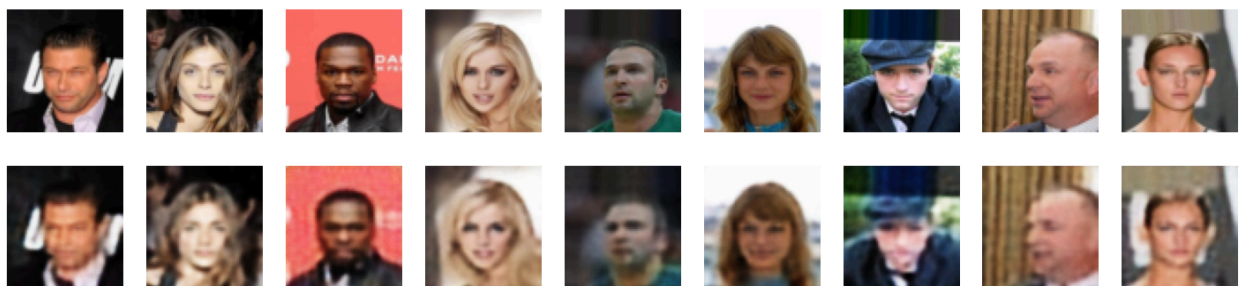
4.1 Training Setup

All models are trained using the Adam optimizer due to its efficiency in handling non-stationary objectives and adaptive learning rates. The learning rate, batch size, and number of training epochs are selected based on computational constraints and model requirements. Training is performed on GPU hardware, such as Kaggle or Google Colab environments, to accelerate computation.

5. Results and Discussion

5.1 Generated Face Samples

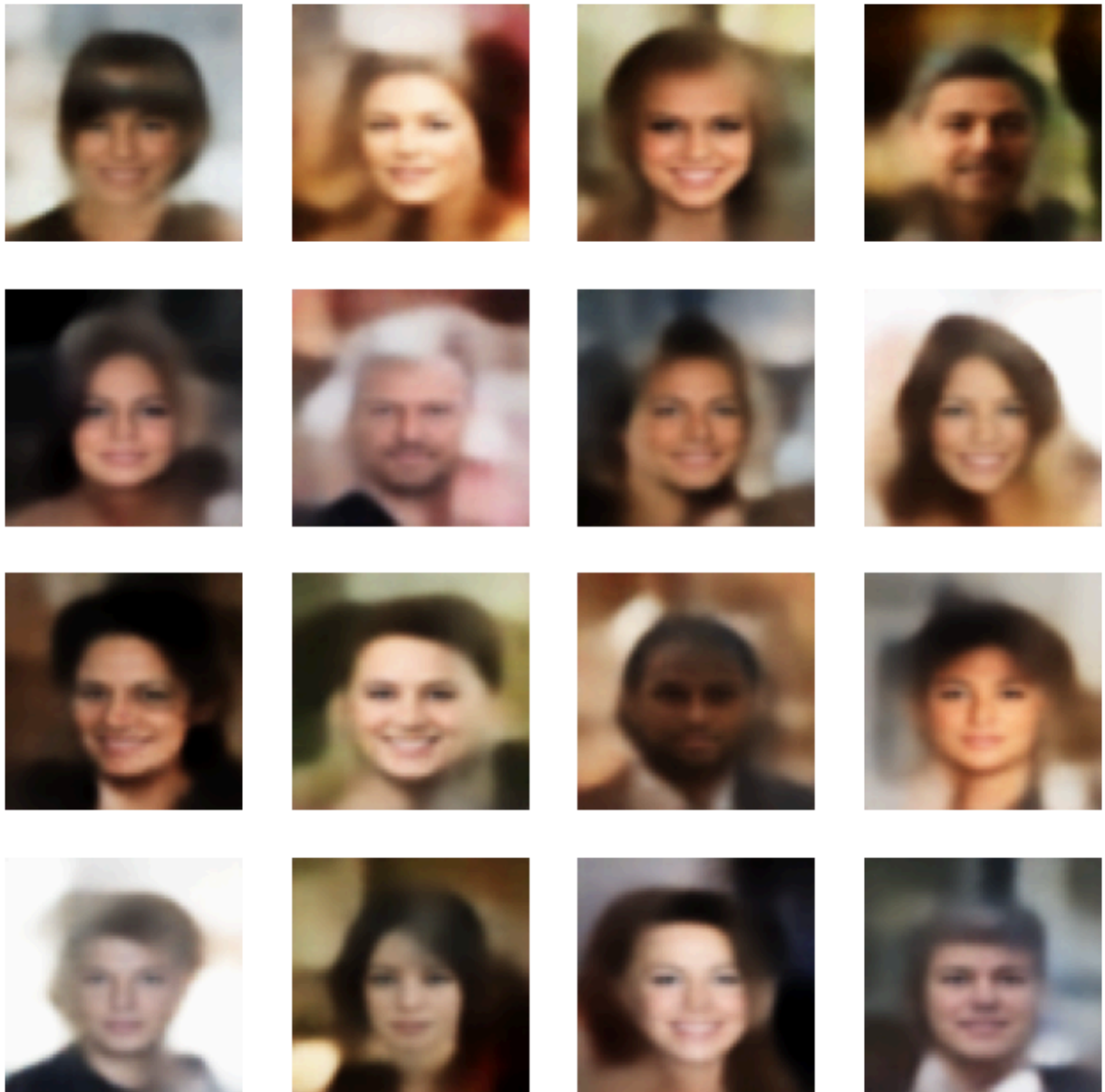
5.1.1 Autoencoder Outputs



The Autoencoder is able to reconstruct input images with reasonable structural accuracy. However, the outputs appear blurry and lack sharp details due to the pixel-wise reconstruction

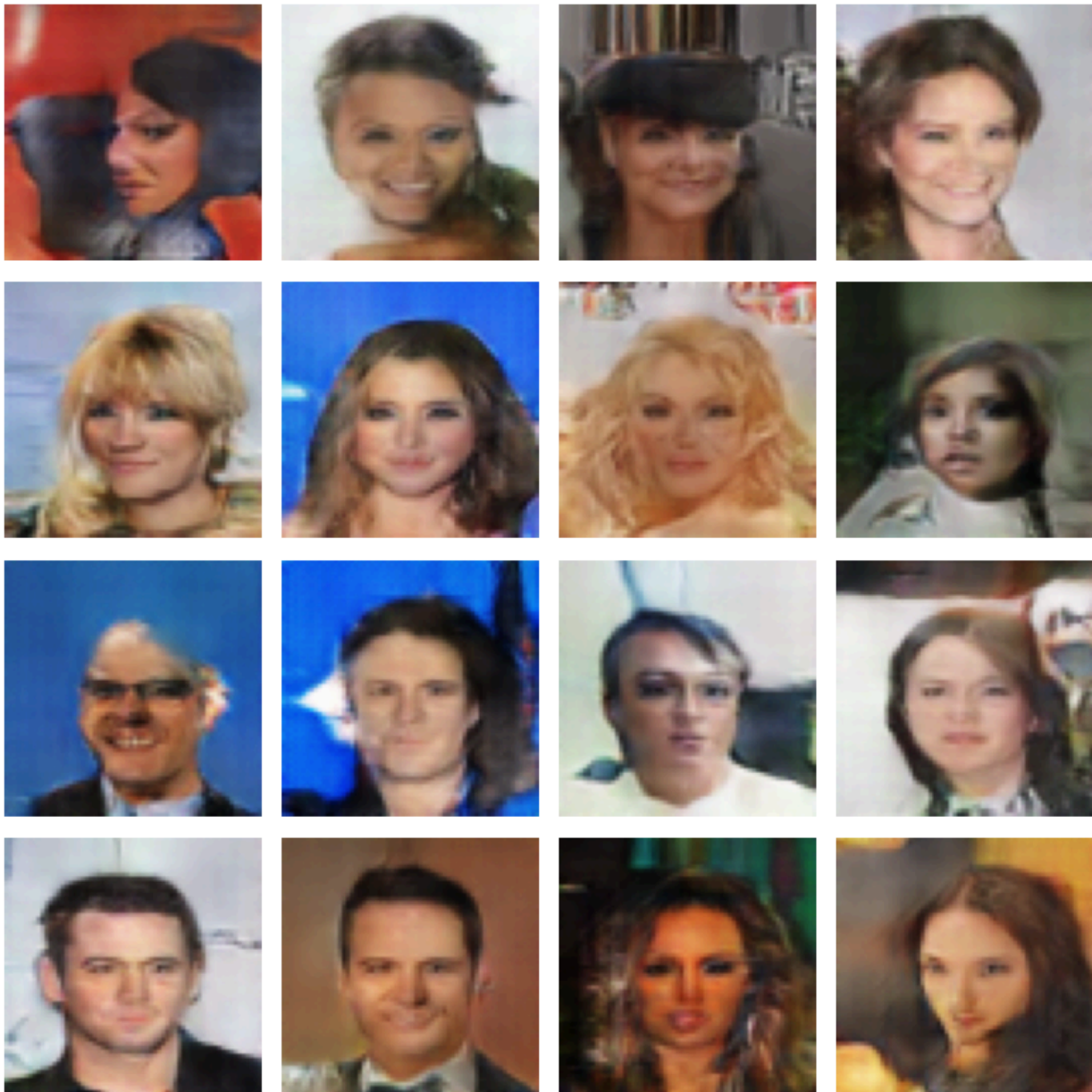
objective. Additionally, the model does not exhibit the ability to generate novel samples, as it primarily learns to replicate input data.

5.1.2 Variational Autoencoder Outputs



The Variational Autoencoder produces more diverse outputs compared to the Autoencoder, owing to its probabilistic latent space. This allows sampling of new images by drawing latent vectors from the learned distribution. However, the images are noticeably blurrier, as the KL divergence term enforces smoothness in the latent space, leading to averaging effects.

5.1.3 GAN / WGAN Outputs



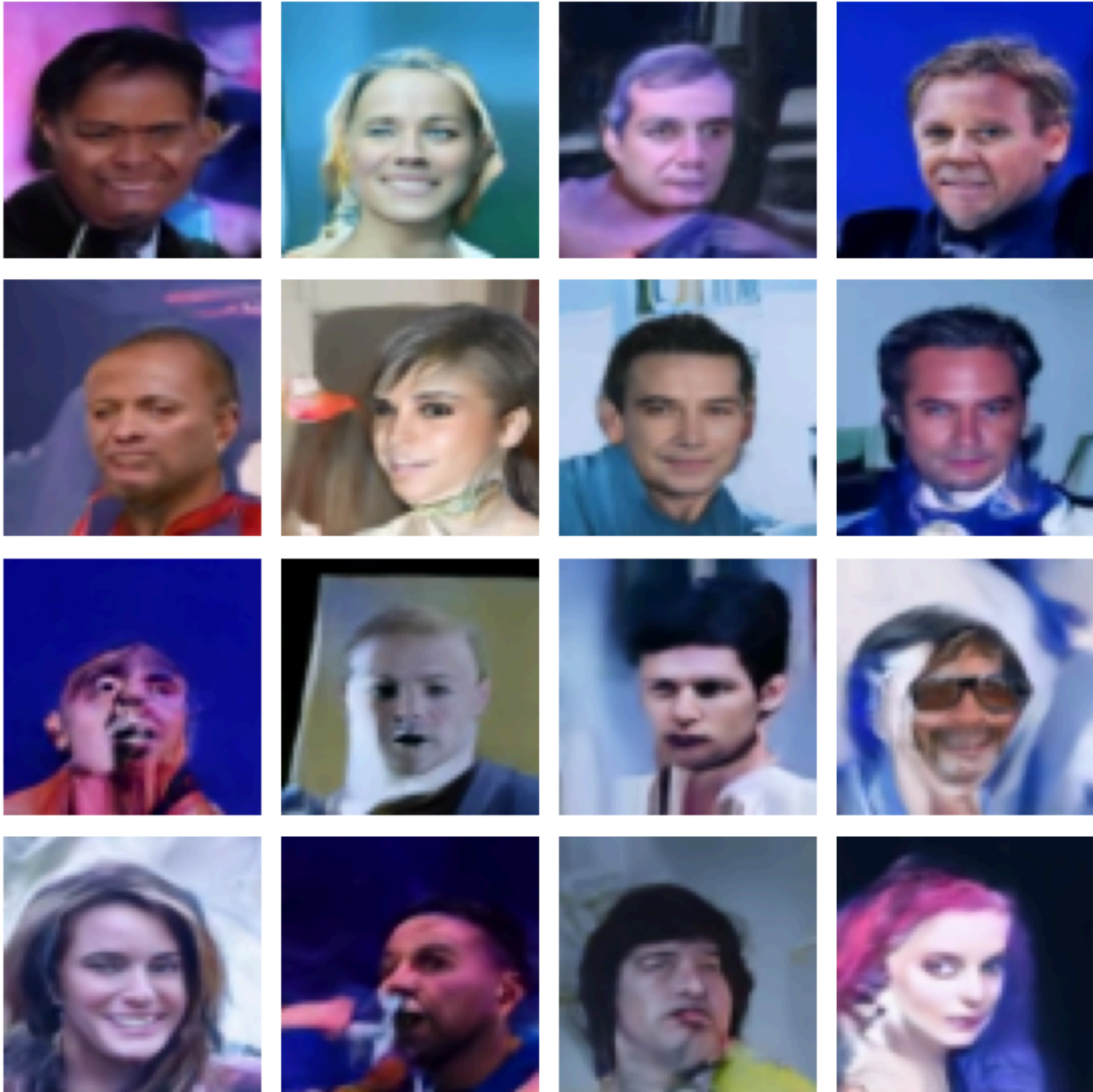
GAN-based models generate significantly sharper and more realistic images. The adversarial training process forces the generator to capture fine-grained details such as textures and edges.

Generated Random Faces



WGAN further improves training stability and reduces issues such as mode collapse, resulting in more consistent outputs.

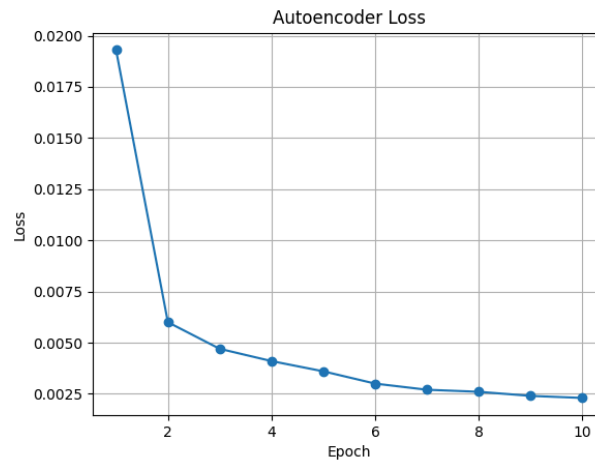
5.1.4 Diffusion Model Outputs



Diffusion models achieve the highest level of realism among all models considered. The generated images exhibit fine details and high perceptual quality. The iterative denoising process enables precise reconstruction of image structures, although it requires higher computational resources and longer training time.

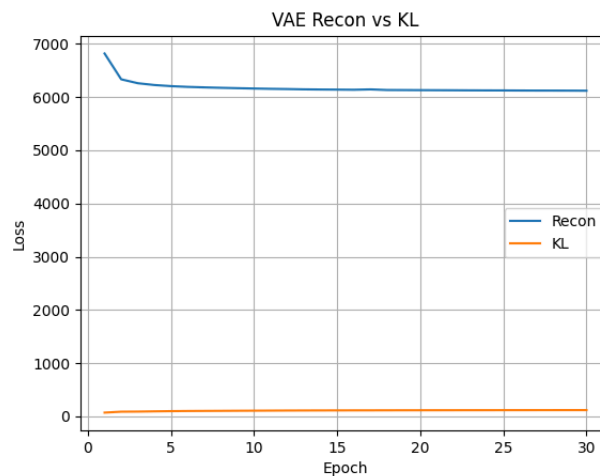
5.2 Training Loss Plots

5.2.1 Autoencoder Loss



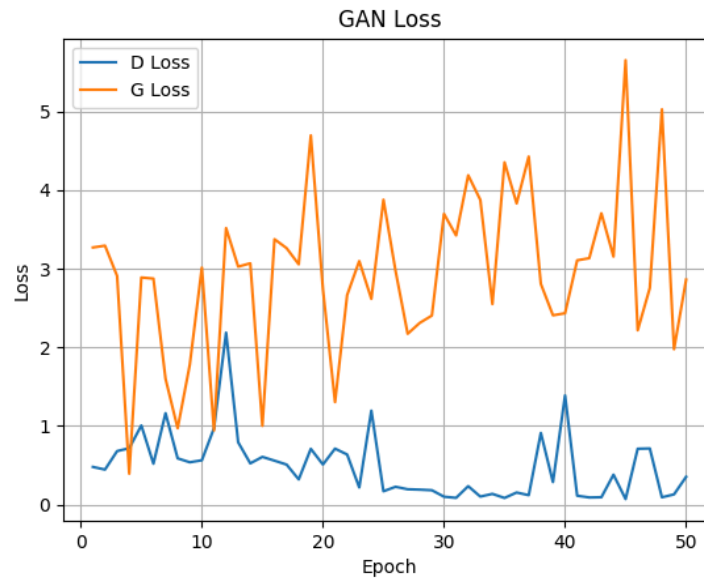
The Autoencoder loss decreases smoothly over training iterations, indicating stable convergence. The model quickly learns to reconstruct input images with minimal error.

5.2.2 VAE Loss

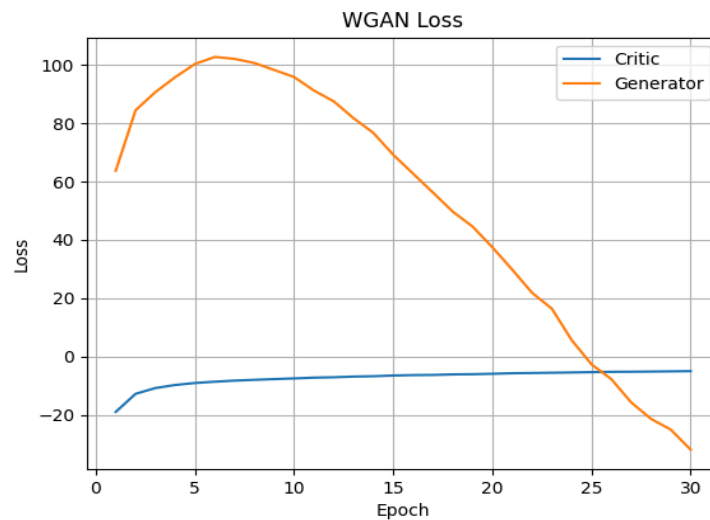


The VAE loss reflects a balance between reconstruction accuracy and latent space regularization. The KL divergence term prevents overfitting but increases overall loss magnitude.

5.2.3 GAN / WGAN Loss

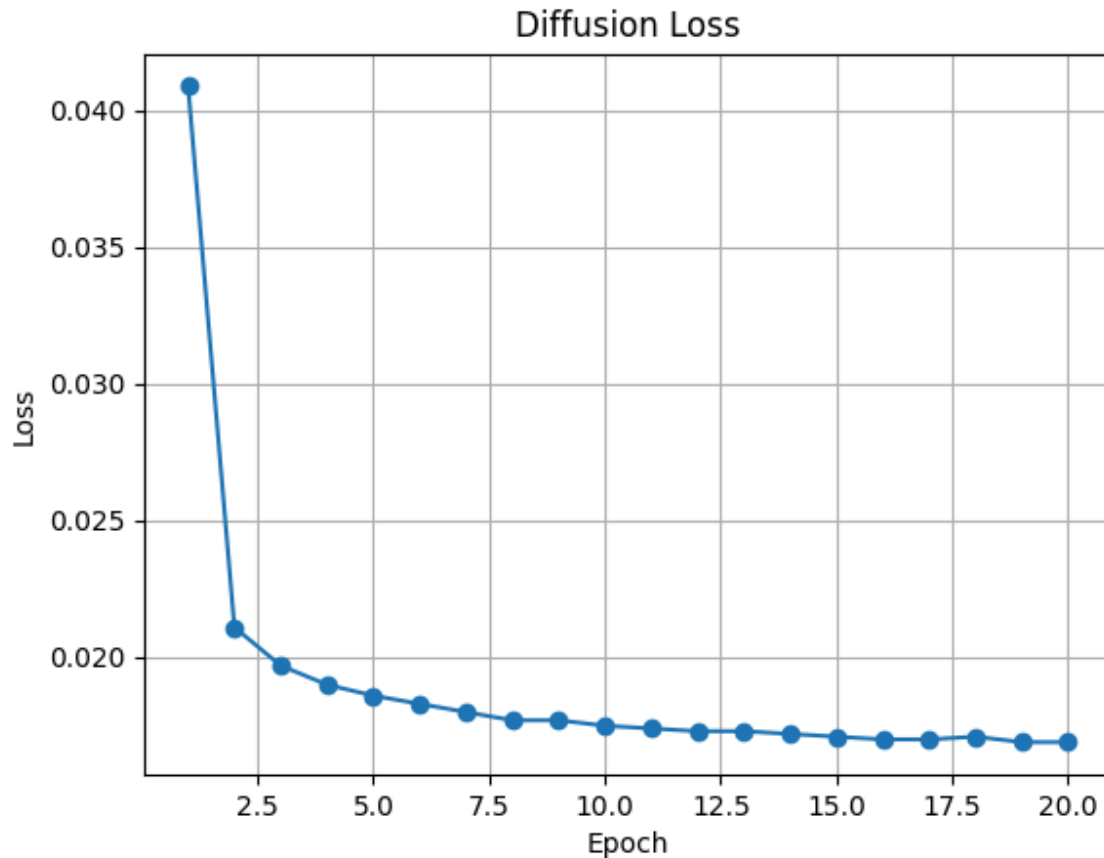


GAN training is characterized by oscillatory loss behavior due to the adversarial interaction between generator and discriminator.



WGAN exhibits smoother and more stable loss curves, indicating improved training dynamics.

5.2.4 Diffusion Model Loss



The diffusion model loss decreases steadily as the network learns to predict noise accurately. Training is stable and does not exhibit oscillations seen in GANs.

5.3 Comparative Analysis

The comparative evaluation of different models reveals distinct trade-offs. Autoencoders are simple and stable but lack generative capability. VAEs introduce diversity but compromise on sharpness. GANs achieve high realism but suffer from instability, while WGAN mitigates some of these issues. Diffusion models outperform all other approaches in terms of image quality and stability, albeit at the cost of increased computational complexity.

Model	Image Quality	Diversity	Training Stability
Autoencoder	Low	Low	High
VAE	Medium	Medium	High
GAN	High	High	Low
WGAN	High	High	Medium
Diffusion	Very High	Very High	High

6. Conclusion

This project demonstrates the evolution of deep generative models for face synthesis. Starting from basic reconstruction-based approaches, the study progresses to advanced generative frameworks capable of producing highly realistic images. Each model contributes unique insights into generative modeling, highlighting the importance of latent space design, training objectives, and optimization strategies.

Diffusion models emerge as the most effective approach for high-quality image generation, while GAN-based models offer a practical balance between performance and efficiency. Future work may involve incorporating evaluation metrics such as Fréchet Inception Distance (FID), exploring hybrid architectures, and optimizing diffusion models for faster inference.

References

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